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EXTENDED ABSTRACT

Evaluating the Potential of Foundation Models for Land Use and Land Cover Classification

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1. Introduction and Objectives

Using Earth Observation data in environmental studies can be challenging due to the multi-modality of the data, strong spatio-temporal dependencies, and massive, largely unlabeled datasets, which limit traditional task-specific models [Mai et al., 2024]. Land Use and Land Cover classification is a common task that serves as a critical benchmark. Recently, Foundation Models (FMs) have gained attention for advantages like multi-modal integration, rapid convergence, and robust generalization, making them an appealing alternative to task-specific models. However, FMs are still relatively new and have not been thoroughly compared for earth observation image classification. This work provides a comparative analysis of FMs and task-specific models for land use and land cover classification, evaluating FMs in a transfer learning setting and examining how architecture, learning method, and pretrained datasets impact performance.

2. Materials and Methods

We established a standardized experimental protocol for comparing FMs under identical conditions. To avoid architecture-specific adaptations, the FMs were not fine-tuned. Instead, embeddings were extracted from pretrained frozen FMs and used for transfer learning with a multilayer perceptron (MLP) classifier, as illustrated in figure 1. The MLP input dimension matches the embedding size, its hidden layer size was selected via hyperparameter search, and its output corresponds to the target classes. As the FMs were not fine-tuned, only the MLP is trained. All models were trained and evaluated on identical data splits. Our study compares three FMs: SatCLIP [Klemmer et al., 2025] is a lightweight dual-encoder FM that aligns geographic coordinates and satellite imagery using contrastive learning. It relies on a geographic location encoding with spherical harmonics and sinusoidal representation networks [Rußwurm et al., 2024] and at inference produces 256-dimensional representations from coordinates alone. Prithvi-EO [Szwarcman et al., 2025] is a large spatiotemporal FM based on a 3D Vision Transformer trained with masked autoencoding, incorporating geolocation and temporal metadata to generate 1280-dimensional embeddings. RemoteCLIP [Liu et al., 2024] is a vision–language FM using a ViT-L image encoder and a transformer text encoder, contrastively pretrained on image–text pairs, producing 768-dimensional embeddings. All models are evaluated on the multispectral EuroSAT dataset [Helber et al., 2017] (27,000 Sentinel-2 images, 64×64) using an identical 60/20/20 data split.

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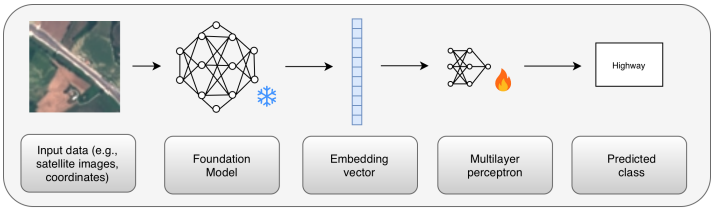

Figure 1. *Evaluation pipeline : feature extraction with frozen FMs followed by MLP classification.*

Table 1. *Key specifications and performance (averaged F1-score over 5-fold cross-validation) of Foundation and task-specific models*

Type of Model	Model	Architecture	Params	Nb Bands	F1-score
Foundation	Prithvi	ViT	600M	6	$0.95 \pm .00$
	RemoteCLIP	ViT+Text	428M	3	$0.76 \pm .01$
	SatCLIP	Swin+ViT	23M	12	$0.60 \pm .01$
Task-Specific	ResNet50	CNN	24M	13	$0.93 \pm .02$
	3-layer CNN	CNN	25K	13	$0.95 \pm .01$

3. Preliminary Results and Discussion

Table1 summarizes the classification performance of FMs and task-specific models on EuroSAT. Prithvi-EO achieves the best performance among FM (averaged F1-score 0.95), likely due to its large and spatiotemporal pretraining. RemoteCLIP ranks second (averaged F1-score 0.76), it is limited to RGB imagery, which might explain the performance gap compared to Prithvi-EO. In contrast, SatCLIP performs significantly less effectively (averaged F1-score 0.60), consistent with its reliance only on geographic coordinates at inference. Overall, the results indicate that FM performance is strongly influenced by pretraining strategy and available input modalities. Models pretrained on large-scale remote sensing data with multispectral information achieve substantially better performance, whereas approaches relying on limited modalities are less effective. This highlights the importance of task-aligned pretraining and multimodal support for downstream remote sensing applications.

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References

[Helber et al., 2017] Helber, P., Bischke, B., Dengel, A., and Borth, D. (2017). EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.

[Klemmer et al., 2025] Klemmer, K., Rolf, E., Robinson, C., Mackey, L., and Rußwurm, M. (2025). SatCLIP: Global, General-Purpose Location Embeddings with Satellite Imagery. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(4):4347–4355.

[Liu et al., 2024] Liu, F., Chen, D., Guan, Z., Zhou, X., Zhu, J., Ye, Q., Fu, L., and Zhou, J. (2024). RemoteCLIP: A Vision Language Foundation Model for Remote Sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 62:1–16.

[Mai et al., 2024] Mai, G., Huang, W., Sun, J., Song, S., Mishra, D., Liu, N., Gao, S., Liu, T., Cong, G., Hu, Y., Cundy, C., Li, Z., Zhu, R., and Lao, N. (2024). On the Opportunities and Challenges of Foundation Models for GeoAI (Vision Paper). *ACM Trans. Spatial Algorithms Syst.*, 10(2):11:1–11:46.

[Rußwurm et al., 2024] Rußwurm, M., Klemmer, K., Rolf, E., Zbinden, R., and Tuia, D. (2024). Geographic location encoding with spherical harmonics and sinusoidal representation networks.

[Szwarcman et al., 2025] Szwarcman, D., Roy, S., Fraccaro, P., Gíslason, Þ. E., Blumenstiel, B., Ghosal, R., Oliveira, P. H. d., Almeida, J. L. d. S., Sedona, R., Kang, Y., Chakraborty, S., Wang, S., Gomes, C., Kumar, A., Truong, M., Godwin, D., Lee, H., Hsu, C.-Y., Asanjan, A. A., Mujeci, B., Shidham, D., Keenan, T., Arevalo, P., Li, W., Alemohammad, H., Olofsson, P., Hain, C., Kennedy, R., Zadrozny, B., Bell, D., Cavallaro, G., Watson, C., Maskey, M., Ramachandran, R., and Moreno, J. B. (2025). Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation Applications. arXiv:2412.02732 [cs].